MACHINE LEARNING LESSON DOCUMENTATION

Feature Selection Methods

Introduction

This report documents my learning from two videos covering the topic of feature selection methods in machine learning. Although the videos were separate, they formed part of the same lesson and helped me understand how feature selection works, why it is important, and how different methods can be applied to a machine learning problem.
I watched the lessons, took notes, and replicated the practical work in my own Python script. This allowed me to follow the concepts explained in the videos and see how they were applied in practice. All the practical work ran successfully.
The main focus of the lesson was to understand how selecting relevant features can affect the training process and performance of a machine learning model. I also learned how to compare different feature selection methods using a classification model and a performance metric.

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Understanding Feature Selection

One of the main things I learned from this part of the lesson is that feature selection is an important step in preparing data for machine learning.
Features are the individual variables or columns used by a model to make predictions. A dataset may contain several features, but not all of them will necessarily be equally useful to the model. Some features may contribute more information than others, while some may provide little useful information for the prediction task.
Feature selection involves identifying and retaining useful features while removing features that may not contribute enough to the model.
I understood that using every available feature is not always the best approach. Depending on the dataset and the problem being solved, reducing the number of features can make the model simpler and more efficient.

Why Feature Selection Is Important

The lesson explained three main reasons for carrying out feature selection.
First, it makes the model easier to maintain. When a model uses fewer features, the information it depends on can be easier to manage. This can also make the overall process more straightforward.
Second, it can reduce training time. A model may need to process less information when unnecessary features are removed. This can make training more efficient, particularly when working with larger datasets.

Third, it can improve model performance. Some features may contain limited useful information for the prediction task. Removing such features may help a model perform better, although this depends on the dataset and the selection method used.
My understanding is that feature selection is not simply about reducing the number of columns in a dataset. It is about identifying a useful set of features that supports the model's objective.

When Feature Selection Should Be Performed

Another important point from the lesson was that feature selection should be carried out before hyperparameter tuning and model selection.
I understood this to mean that the features used to train a model should be considered as part of the model-development process, rather than treated as an afterthought.
The selection process can affect the information available to the model, so it is useful to examine the features before deciding how to tune and compare models.

Loading and Understanding the Dataset

For the practical work, the lesson used the Wine dataset available through Scikit-learn.
I replicated the process of loading the dataset and converting its feature data into a Pandas DataFrame. The dataset contains measurements associated with different wine samples. These measurements include alcohol, malic acid, ash, magnesium, total phenols, flavonoids, colour intensity, hue, and proline, among other features.
A target column was also added to the DataFrame. This column identifies the class associated with each sample and provides the information the classification model is expected to predict.
Using Pandas made it easier to view the data in a tabular format and inspect the different columns. I also examined the first few rows to understand how the dataset was organised.
This practical exercise reinforced the importance of understanding the data before proceeding to model training. It is necessary to know which columns contain the input features and which column represents the target.

Exploring the Data Visually

The lesson also demonstrated how visualisation can help with understanding a dataset.
I replicated the use of a Seaborn swarm plot to examine selected features in relation to the target classes. The features included alcohol, malic acid, and alkalinity of ash.
The plot displayed the feature values and used the target classes to distinguish the observations. This made it possible to examine how the values were distributed across the different classes.

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I understood that visualisation provides an additional way of examining the data before training a model. Instead of relying only on tables of numbers, I could use a graph to observe patterns and differences between features.
The lesson also involved examining the number of observations belonging to each target class. This helped demonstrate how the dataset was distributed among its classes.

Dividing the Dataset into Training and Testing Data

Before training the model, the dataset was divided into training and testing sets using Scikit-learn's train_test_split function.
The input features were separated from the target column. The features were assigned to X, while the target was assigned to y.
The data was then split into training and testing portions. The training data was used to fit the model, while the testing data was reserved for evaluating its predictions.
I understood that this separation is important because a model should be evaluated using data that was not used to fit it. This gives a more useful indication of how it performs on unseen observations.
The notebook also used a fixed random state in the practical workflow. I understood that setting a random state helps make the split reproducible, allowing the same split to be obtained when the code is run again under the same conditions.

Establishing a Baseline Model

Before applying feature selection, the lesson established a baseline using a Gradient Boosting Classifier trained with all the available features.
A baseline provides a reference point against which later approaches can be compared. In this case, the model was trained using the original feature set so that its performance could be compared with the results obtained after selecting features.
The notebook used Scikit-learn's GradientBoostingClassifier and evaluated its predictions using the weighted F1-score.
The recorded baseline weighted F1-score was 0.908.
This result provided a starting point for evaluating whether the feature selection methods could achieve comparable or better performance using fewer features.
I learned that it is useful to establish a baseline before making changes to a model's inputs. Without a baseline, it would be more difficult to determine whether a feature selection method had helped or reduced performance.

Variance Threshold as a Feature Selection Method

The first feature selection method explored in the practical work was based on variance.

Variance measures how much the values of a feature differ from their mean. A feature with very low variance has values that change very little across the observations, while a feature with higher variance has values that are more spread out.
The lesson demonstrated how to calculate the variance of the training features and examine the results.
However, comparing raw variances requires care because features may use different measurement scales. A feature measured in larger numerical units can have a higher variance simply because of its scale.
To examine this issue, the practical work used MinMaxScaler to scale the training features. The scaled feature values were then used to inspect the differences in variance.
The lesson subsequently demonstrated removing the ash and magnesium features and training the classifier with the remaining features.
The resulting weighted F1-score was compared with the baseline score.
My main understanding of this method is that variance can be used as a simple way to identify features with little variation. However, low variance does not automatically mean a feature is useless for prediction. A feature's usefulness also depends on its relationship with the target.
For this reason, variance-based selection should be evaluated rather than assumed to improve performance in every situation.

My Observations from Version 1

This part of the lesson helped me understand the stages that come before and during feature selection. I learned that it is important to inspect the dataset, identify the target, divide the data appropriately, and establish a baseline before evaluating different feature selection methods.
I also gained a better understanding of why data visualisation and scaling can be useful when examining features.
The most important lesson for me was that removing features should be a considered decision. A feature should not be removed simply because its values appear small or because it has a lower variance than other features. Its usefulness should be considered in relation to the prediction task and the model's performance.

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Moving Beyond Variance-Based Selection, The practical work continued with other feature selection methods that could be compared with the variance-based approach.
I understood that feature selection can be approached in different ways. Some methods examine statistical properties of the features, while others use information about the target or evaluate features through a model.

Using different methods is useful because a feature that appears less useful under one approach may still contribute to the model's predictive performance under another.
The lesson demonstrated three additional methods: SelectKBest using mutual information, Recursive Feature Elimination (RFE), and Boruta.

SelectKBest and Mutual Information

The next method demonstrated was SelectKBest, used with mutual_info_classif.
SelectKBest selects a specified number of features based on the scores calculated by a scoring function. In this practical exercise, mutual information was used to assess the relationship between each feature and the target.
I understood mutual information as a way of estimating how much information one variable provides about another. In this case, it was used to assess how informative individual features were about the target classes.
The practical work tested different values of k, where k represented the number of features to select.
For each value of k, the selected features were used to train the Gradient Boosting Classifier. The model's predictions were then evaluated using the weighted F1-score.
The scores were collected and plotted so that performance could be compared across different numbers of selected features.
This approach helped me understand that the number of features retained can influence model performance. Selecting too few features may remove useful information, while retaining more features does not automatically guarantee better results.
The aim is to find a suitable number of features for the task, using evaluation results to guide the decision.

Recursive Feature Elimination (RFE)

The next method covered was Recursive Feature Elimination, commonly referred to as RFE.
Unlike a simple approach that ranks features individually and selects a fixed number, RFE uses a model-based process to help determine which features to retain.
The method works by fitting an estimator, ranking features according to its importance mechanism, and removing less important features. The process is repeated until the required number of features remains.
In the practical work, RFE was applied to the training data. The selected features were then used to train the Gradient Boosting Classifier, and the model was evaluated using the weighted F1-score.
The notebook also examined how the score changed as the number of retained features changed. This made it possible to compare different feature-set sizes rather than relying on only one selection result.

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I learned that RFE provides a systematic way of reducing the number of features while considering the model used in the selection process.
It also reinforced the importance of evaluating the selected features. The purpose is not simply to remove as many columns as possible, but to find a feature set that gives a useful balance between simplicity and predictive performance.

Boruta Feature Selection

Boruta was another method demonstrated in the practical work.
Boruta is a feature selection method designed to identify relevant features by comparing the importance of original features with that of randomly generated reference features, often called shadow features.
This comparison helps the method assess whether the original features show meaningful importance beyond what might be expected from randomised data.
The notebook applied the Boruta selector to the training features and target. The selected features were then used to train the Gradient Boosting Classifier, and the resulting predictions were evaluated.
The notebook recorded a Boruta weighted F1-score of 0.982.
This result formed part of the comparison between the different approaches explored in the lesson.
My understanding is that Boruta offers a different way of assessing feature relevance from the methods based on variance or mutual information. It provides a feature-selection approach that considers feature importance in relation to randomly generated reference features.
However, I also learned that a strong score from one experiment does not mean the same method will always perform best on every dataset. The result depends on the dataset, the model, the selection settings, and the evaluation process.

Evaluating the Models with the Weighted F1-Score

Throughout the practical work, the weighted F1-score was used to evaluate and compare the classifier's predictions.
The F1-score combines precision and recall into a single measure. Precision considers how many of the observations predicted as a particular class were correctly identified, while recall considers how many of the actual observations belonging to that class were identified correctly.
The weighted F1-score calculates the F1-score for each class and combines the results while accounting for the number of observations in each class.
I understood that this metric was useful for comparing the classification results produced by the different feature selection methods.

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The practical work used the same general classifier and compared its performance with different feature sets. This helped make the comparison more meaningful because the feature selection approach was the main factor being examined.
It also showed me why a model's performance should be measured rather than judged only by how much the number of features has been reduced.

Comparing the Feature Selection Methods

The notebook brought the results together in a visual comparison of the weighted F1-scores.
The approaches examined included the baseline model using all features, variance-based selection, SelectKBest, RFE, and Boruta.
The comparison helped demonstrate that feature selection is an experimental process. Different methods may select different features, and the resulting models may perform differently.
I learned that it is important to consider both the number of features retained and the model's performance. A smaller feature set may be attractive because it reduces complexity, but it should still provide acceptable predictive results.
Similarly, a method that achieves a high score in one experiment should not automatically be considered the best choice for every problem.
The most suitable method depends on the objective of the work and the results obtained during evaluation.

My Observations from Version 2

This part of the lesson expanded my understanding of feature selection by showing that there is more than one way to identify useful features.
SelectKBest helped me understand how features can be ranked using a scoring function and how the number of selected features can be varied. RFE demonstrated a model-based process of progressively removing features. Boruta introduced another approach that evaluates feature relevance by comparison with randomly generated reference features.
I also understood the importance of using a consistent evaluation metric when comparing the results of different methods.
The main lesson I took from this section is that feature selection requires testing and comparison. It is not enough to select a method based on its name or theoretical advantages. The method should be assessed using the particular dataset and prediction task.

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COMBINED REFLECTION ON BOTH VERSIONS

What I Learned Overall

The two videos helped me build a more complete understanding of feature selection in machine learning, from preparing and examining the data to applying different selection methods and evaluating the resulting models.

I learned that feature selection can help simplify a model, reduce training requirements, and potentially improve performance. I also learned that these benefits are not automatic and should be verified through evaluation.

The practical work gave me the opportunity to apply the concepts using Python, Pandas, Matplotlib, Seaborn, and Scikit-learn. I followed the process of loading the Wine dataset, examining the features, separating the input variables from the target, splitting the data into training and testing sets, and establishing a baseline model.

I then replicated the different feature selection approaches and compared their results using the weighted F1-score.

My Approach to the Practical Work

My approach was to watch the lesson videos, take notes, and then replicate the practical work in my own Python script.

Following the process in this way helped me connect the explanations in the lesson with the actual code. I could see how the dataset was prepared, how the methods were applied, and how the model's performance was evaluated.

All the practical work ran successfully. Replicating it also gave me an opportunity to become more familiar with the libraries and functions used in the lesson.

I found that working through the code was useful because it helped me understand how the different stages fit together rather than treating each function as an isolated instruction.

What I Found Most Useful

The most useful aspect of the lesson was seeing the different feature selection methods applied to the same dataset and evaluated using a common metric.

This made it easier to understand why feature selection should be treated as part of the model-development process.

I also found the use of visualisations helpful. The plots made it easier to examine feature distributions and compare model scores.

Another useful point was learning to establish a baseline before applying feature selection. This provided a reference against which the other methods could be assessed.

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What I Found Interesting

I found it interesting that feature selection can be approached in several different ways. Variance-based selection examines how much the values of a feature vary. SelectKBest with mutual information considers the information a feature provides about the target. RFE uses a model-based elimination process, while Boruta compares feature importance against randomly generated reference features.

Although these methods share the same broad objective, they approach the task differently. This helped me appreciate why it is useful to understand the method being applied instead of treating feature selection as one single technique.

I also found it interesting that the model's performance needed to be checked after the features were selected. Reducing the number of features was not, by itself, enough to establish that the selection had been successful.

Challenges and Areas Requiring Care

One area that required careful understanding was the relationship between feature variance and feature usefulness.

A feature with low variance is not necessarily irrelevant to the target. Likewise, a feature with a high variance is not automatically useful for classification. Variance can help identify features worth examining, but it does not independently establish their predictive value.

I also recognised the importance of scaling when comparing features with different numerical ranges. Scaling can make the values easier to compare on a common range, although it does not make every feature equally useful.

Another area requiring care was interpreting the weighted F1-score. A score provides information about model performance, but it should be considered alongside the task being solved and the way the model was evaluated.

These points reminded me that machine learning involves more than running code. It also requires understanding what the results mean and making decisions based on the evidence.

Connection to My Previous Learning

This lesson connects with the wider machine learning workflow because it brings together data preparation, exploratory analysis, training and testing, model fitting, prediction, and evaluation.

It reinforced the importance of separating input features from the target variable and keeping training and testing data distinct.

It also showed how Python libraries can be used together to complete different parts of the workflow. Pandas was used to organise the data, visualisation libraries helped

examine it, and Scikit-learn provided tools for splitting the data, selecting features, training the classifier, and evaluating predictions.
The lesson therefore helped me see how the individual techniques I learn fit into a complete practical process.

Conclusion

Overall, the two videos improved my understanding of feature selection methods and their role in machine learning.
I learned how to prepare a dataset, establish a baseline model, apply different feature selection methods, and compare the resulting classification performance using the weighted F1-score.
By taking notes and replicating the practical work in my own Python script, I was able to reinforce the concepts covered in the lessons and observe how they were applied.
My main takeaway is that feature selection should be guided by the purpose of the model and supported by evaluation. The objective is not simply to use fewer features, but to identify a useful set of features that provides an appropriate balance between model simplicity and predictive performance.